

Applied AI in Biomedicine - Final Assignment Report

Lung Cancer Classification

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1 Introduction

In this project, we tackle *image classification* in the context of lung cancer, the deadliest malignancy worldwide (responsible for 18% of total cancer deaths in 2020), and often diagnosed at advanced stages with a low 5-year survival rate. To address the growing need for accurate screening and detection, we employ **deep learning** techniques on a dataset of CT scan slices, each annotated with a malignancy score (from 1 to 5). We develop four neural networks: two trained on *full-slice* images and two on *zoomed nodule* images, each tasked with both a **binary** (benign vs. malignant) and a **5-class** (score 1–5) classification. Each model also provides a confidence estimate for its predictions. The dataset, although challenging and unbalanced, comprises approximately 2,300 patients, each with two CT images (full slice and zoomed slice) and a corresponding malignancy label.

2 Dataset

The dataset consists of approximately **2300 patients**, each of whom underwent a **CT scan** for lung nodule analysis. From each scan, **only the slice containing the largest nodule area was selected**, reducing data complexity while retaining critical information about the lesion.

For each patient, there are **two CT images** in NRRD format:

- A **full slice** of the chest region, including the nodule (`pat1_fullslice.nrrd`).

- A **zoomed slice**, focused solely on the nodule (`pat1_nodule.nrrd`).

Additionally, an **Excel file** (`dataset_lung.xlsx`) provides a **malignancy score** for each patient, ranging from **1 to 5**, which classifies the tumor type. To facilitate binary classification, we introduced an additional **TumorClass label**, where:

- Patients with **TumorClass 1, 2, and 3** were assigned **0 (benign)**.
- Patients with **TumorClass 4 and 5** were assigned **1 (malignant)**.

2.1 Challenges in the Dataset

1. Severe class imbalance:

- The dataset is **highly imbalanced** across the **five tumor classes**, with **class 3 being significantly overrepresented** compared to the others.
- When performing **binary classification (benign vs. malignant)**, the dataset is also **heavily skewed towards the benign class**.

2. Variable image sizes in the nodule slices:

- Each `pat1_nrrd` image has a **different resolution**, making it challenging to standardize input dimensions for deep learning models.

These issues required **specific preprocessing techniques** to **balance the dataset** and **normalize image dimensions** before training the models.

We started by exploring the dataset to analyze its composition. To do so, we visualized 25 full-slice images, selecting 5 from each tumor class (figure 1). We did the same with the zoomed nodule images (figure 2).

Afterwards, we investigated the cancer distribution among the five tumor classes (figure 3). From these graphs, it is clear that the dataset is widely unbalanced—with 46.21% of the samples in tumor class 3, 19.33% in label 2, 17.68% in label 4, 10.32%

in label 1, and 6.23% in label 5. This imbalance led us to apply preprocessing operations to address the issue.

For binary classification, we added a new target column indicating whether the cancer is benign or malignant. Images with labels 1, 2, and 3 were assigned 0 (benign), while those with labels 4 and 5 were assigned 1 (malignant). Figure 4 shows that 75.9% of the samples are benign versus 24.1% malignant.

Furthermore, to better illustrate the overall data distribution, figure 5 compares the distributions for the multiclass and binary classification problems.

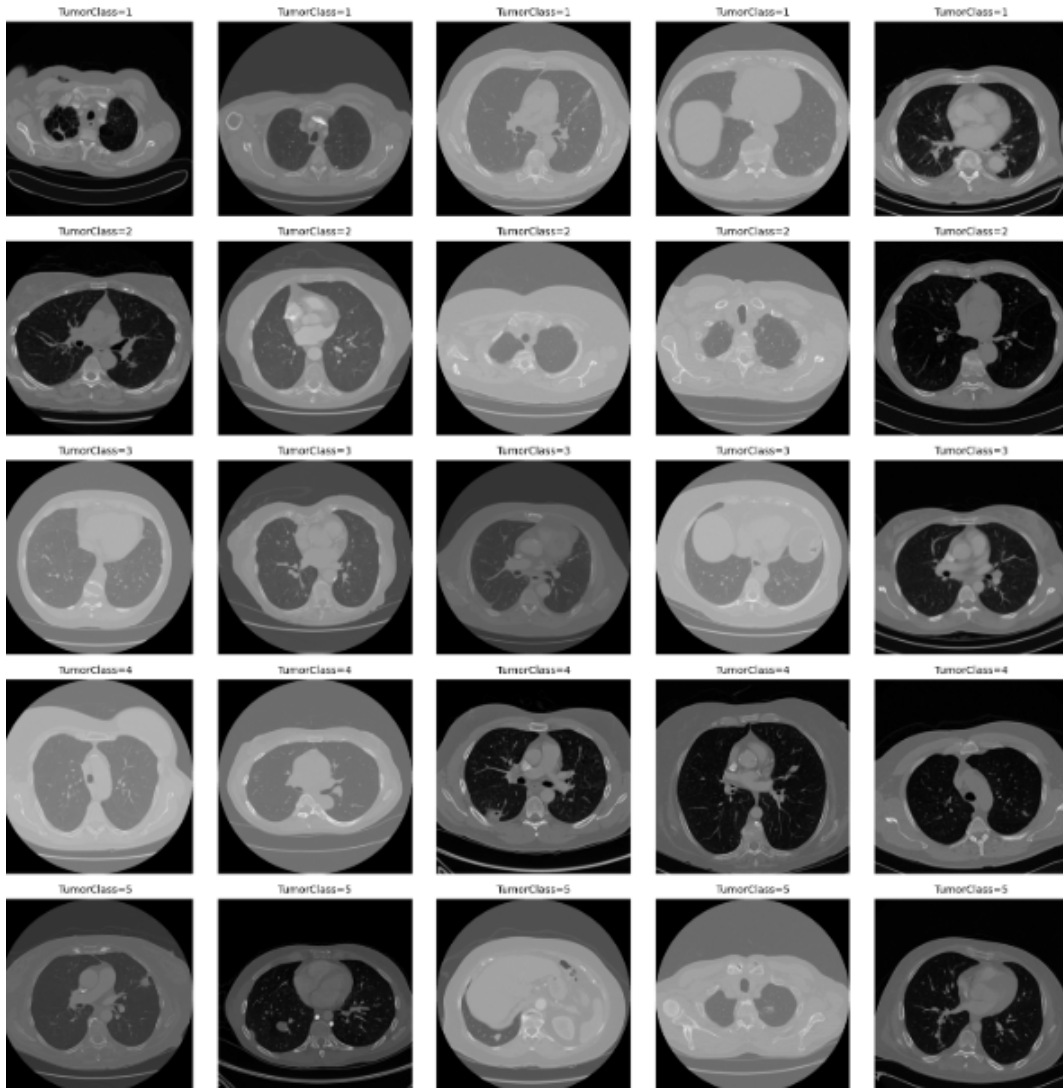


Figure 1: 25 full-slice images, selecting 5 from each tumor class within our dataset.

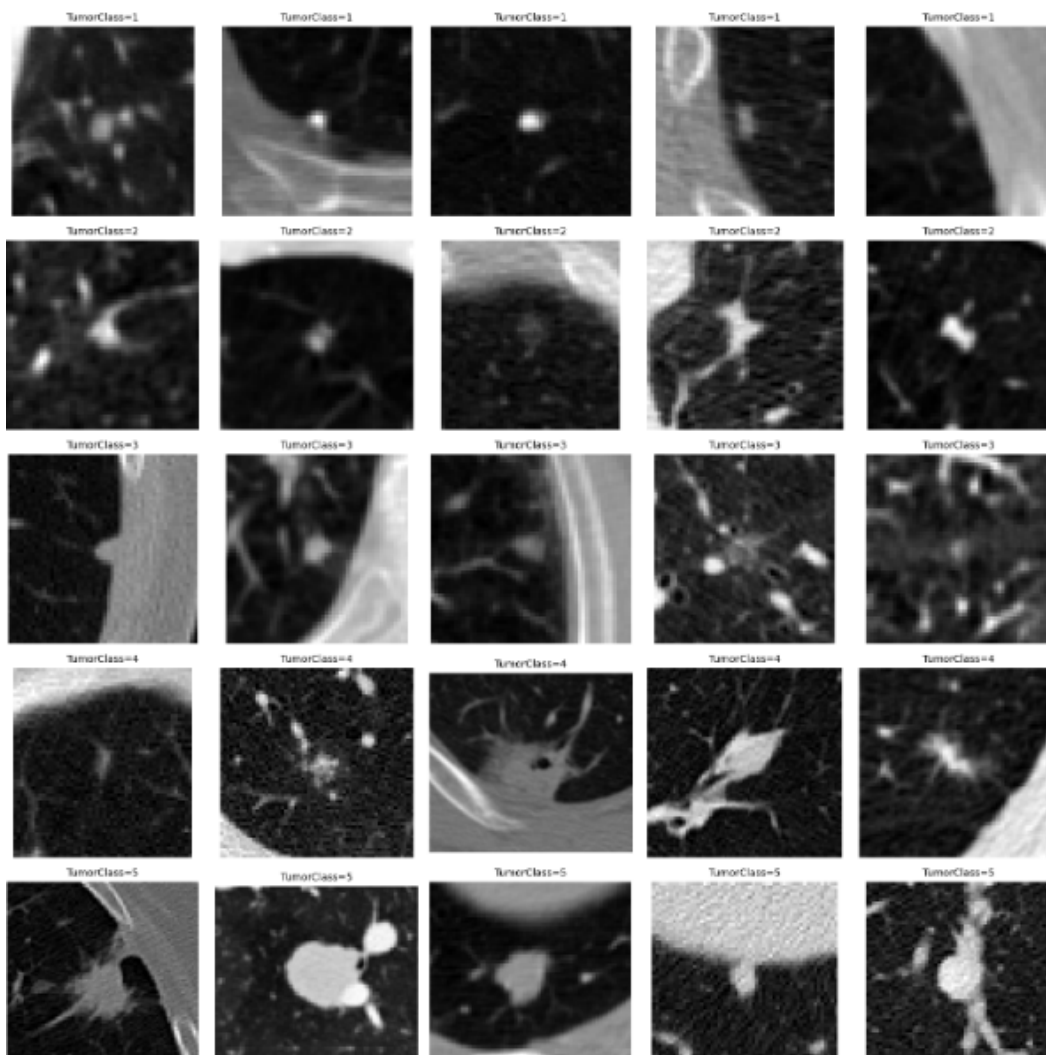


Figure 2: 25 zoomed nodule images, selecting 5 from each tumor class within our dataset.

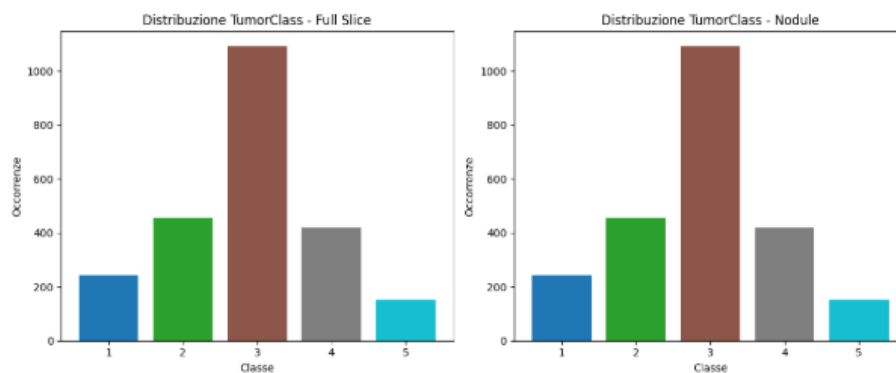


Figure 3: Distribution of tumor classes within the dataset.

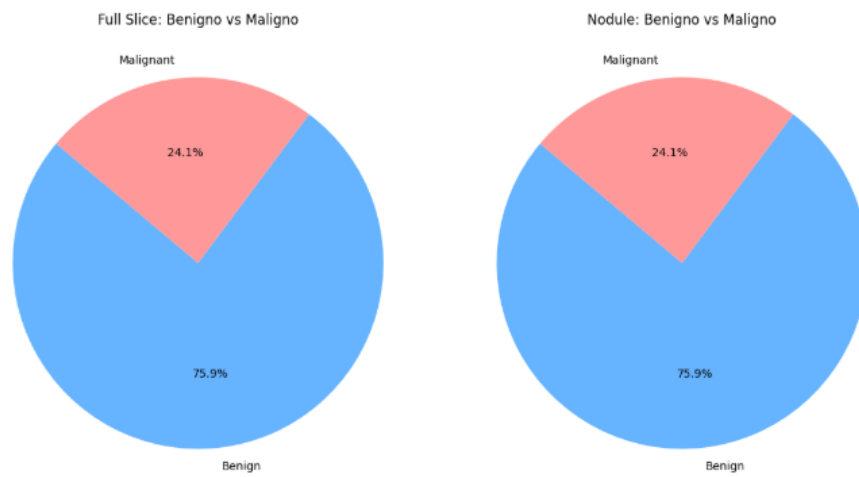


Figure 4: Class distribution among 'Benign' and 'Malignant' labels.

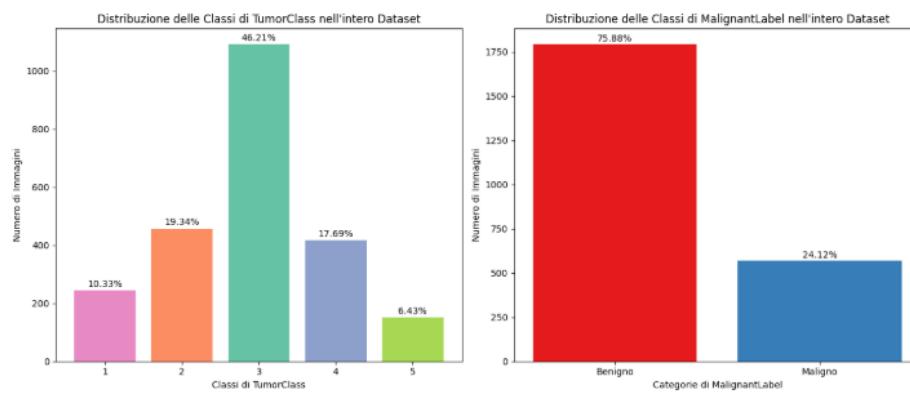


Figure 5: Distribution of samples for multiclass and binary classification problems.

3 Preprocessing

Preprocessing plays a fundamental role in preparing medical imaging datasets for deep learning classification models. This stage ensures that data is standardized, balanced, and efficiently structured before being fed into convolutional neural networks (CNNs). By applying robust preprocessing techniques, we improve model generalization, reduce noise, and enhance feature extraction. While different models require slight variations in preprocessing techniques, there are several common steps that remain consistent across both binary and multiclass classification tasks.

3.1 Common Preprocessing Steps

The preprocessing pipeline begins with image loading and transformation. Lung nodule images, provided in the **NRRD** format, are first converted into a standard 2D representation for compatibility with CNN architectures. Since the raw images are grayscale, they are transformed into three-channel RGB format by replicating the grayscale information across all three channels. This ensures compatibility with pre-trained architectures designed for RGB inputs.

To ensure a uniform input format, all images are resized to a fixed resolution. A resolution of **128×128 pixels** is used for zoomed nodule images, optimizing computational efficiency while maintaining sufficient detail. Conversely, full-slice images require a larger resolution to preserve anatomical structures, as the tumor occupies a small portion of the scan.

Additionally, dataset optimization techniques are applied to enhance computational efficiency. Mini-batch processing (batch size = 8) is implemented to stabilize training and prevent overfitting. Prefetching mechanisms ensure continuous data loading, reducing memory bottlenecks and speeding up the training process.

3.2 Differences in Preprocessing Between Models

While the preprocessing framework shares fundamental steps across all models, certain adaptations are necessary depending on the classification task. These differences arise due to variations in dataset

structure, class distributions, and learning objectives:

- **Class Labeling and Balancing:**

- In **binary classification**, tumor classes are grouped into two labels: benign (Classes 1, 2, 3) and malignant (Classes 4, 5). This simplifies the learning task but may obscure finer distinctions between tumor types.
- In **multiclass classification**, each tumor type retains its individual class label (0 to 4), demanding a more granular decision-making process from the model. This approach provides richer diagnostic insights but increases classification complexity.

- **Class Weighting:**

- Class imbalance is a common issue in medical imaging datasets, requiring weight adjustments to prevent the model from favoring majority classes.
- In binary classification, a simple class weight formula is used to counteract imbalanced distributions:

$$w_c = \frac{1}{n_c} \times \frac{N}{2}$$

where:

- * w_c is the weight assigned to class c ,
 - * n_c is the number of samples in class c ,
 - * N is the total number of samples in the dataset.
- In multiclass classification, a normalized weighting strategy is applied to ensure a fair contribution of each class:

$$w_c = \frac{N}{n_c \cdot \sum_{c=0}^{C-1} \frac{N}{n_c}}$$

where:

- * C is the total number of classes,
- * The denominator represents the sum of inverse class frequencies,
- * Underrepresented classes receive a higher weight, making the model more sensitive to rare occurrences.

- These strategies ensure that the loss function appropriately penalizes errors in underrepresented classes, improving model robustness.

- **Data Augmentation:**

- Data augmentation is selectively applied to enrich the training set and improve model generalization.
- **Binary classification models** do not employ augmentation, as preserving original diagnostic features is paramount.
- **Multiclass classification models** utilize targeted augmentation techniques, such as rotation and translation, to mitigate class imbalance and enhance feature diversity.
- Various advanced augmentation strategies were tested, including:
 - * **FourierMix:** Applied frequency-based transformations to blend images, aiming to improve generalization while preserving relevant spatial structures.
 - * **AugMix:** Introduced stochastic combinations of multiple augmentations to create diverse but realistic variations of the training images.
 - * **RandAugment with GridMask:** Implemented randomized augmentation policies combined with occlusion-based augmentations to enhance feature extraction robustness.
- However, after extensive testing, it was observed that these methods did not yield significant improvements in model performance. In some cases, they introduced distortions that reduced the model’s ability to learn relevant diagnostic features.
- Consequently, only selected augmentations such as controlled rotation and translation were retained for multiclass classification models, ensuring an optimal balance between data enrichment and feature preservation. (see Figure 6)

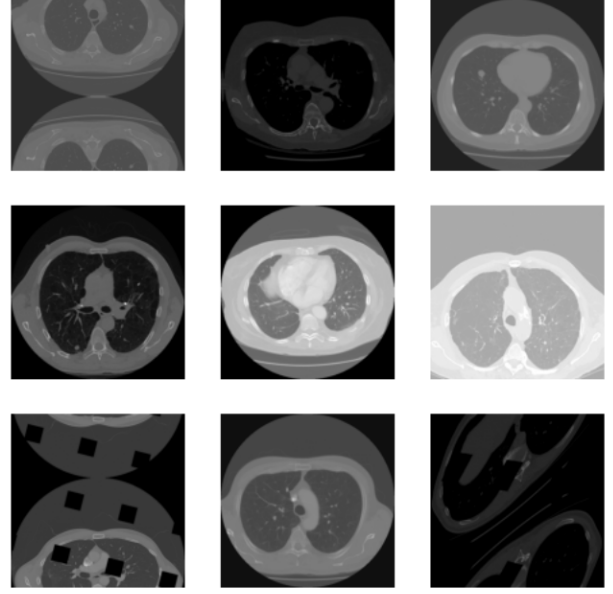


Figure 6: Tested Augmentations

- **Dataset Splitting:**

- Both classification models follow an **80-10-10** split for training, validation, and testing to ensure reliable performance assessment.
- **Stratified sampling** is employed to maintain class distribution consistency across splits, reducing the risk of biased training.
- The validation set plays a crucial role in hyperparameter tuning, ensuring that the model generalizes well before being tested on unseen data.

By integrating these preprocessing techniques, both binary and multiclass classification models are optimized for learning meaningful patterns while addressing dataset limitations such as class imbalance, input variability, and computational constraints. A well-structured preprocessing pipeline ensures model consistency while allowing task-specific modifications to maximize predictive performance.

4 Method

In this section of our report, we present our four main models chosen for their confident estimates, outlining their architectures and training configurations. In the following section, we detail additional experiments that guided our final model choices.

4.1 Zoomed Nodule Binary Classification Model and Training

In this work, we developed a DenseNet121-based transfer learning model to perform binary classification on zoomed nodule images extracted from CT scans, aiming to distinguish between benign and malignant tumors.

Data Preparation: We use the preprocessed, zoomed CT nodule images described in Section *Preprocessing*, each resized to 128×128 pixels and converted to three-channel RGB for compatibility with DenseNet121.

Model Architecture: We use DenseNet121 as the backbone, leveraging ImageNet-pretrained weights but removing its top classification layers. Concretely:

- **Base Model:** DenseNet121 (ImageNet weights), with the top layers removed.
- **Fine-Tuning:** Layers up to index 100 remain frozen, retaining low-level features; subsequent layers are unfrozen for domain-specific adaptation.
- **Classifier Head:** Global Average Pooling reduces spatial dimensions, followed by a Dense layer (128 neurons, ReLU), Dropout (0.3), and a final sigmoid neuron for the binary output.

The model is compiled using Adam and binary cross-entropy loss.

Training Strategy:

- **Dataset Splitting:** We follow the 80-10-10 splitting approach from Section *Preprocessing*, further refined with stratified sampling.
- **Class Balancing:** To address benign-malignant imbalance, we apply the binary

weighting scheme also explained in Section *Preprocessing*.

- **Callbacks:** Early stopping and learning rate reduction on plateau are used to prevent overfitting.
- **Cross-Validation:** A 5-fold stratified cross-validation is performed on the combined training+validation set to fine-tune hyperparameters and estimate performance robustness.

Rationale: This combination of transfer learning from ImageNet, partial layer freezing, and class weighting is designed to stabilize training on a limited dataset and place more emphasis on malignant nodules. Early stopping and learning rate scheduling help avoid overfitting, leading to a model that generalizes effectively to unseen data.

Figure 9 shows the test-set confusion matrix, achieving an accuracy of 0.83 ± 0.034 . Figure 10 reports Precision, Recall, and F1 Score on the same set. The ROC curve (Figure 11) yields an AUC of 0.89, and the Precision-Recall curve (Figure 12) results in an Average Precision (AP) of 0.75%.

4.2 Zoomed Nodule Multiclass Classification Model and Training

For multi-class classification of zoomed nodule images, we employ a MobileNetV2-based model due to its balance between computational efficiency and strong performance.

4.2.1 Model Architecture

MobileNetV2 (pre-trained on ImageNet) serves as a feature extractor, leveraging depthwise separable convolutions for efficient representation learning. The convolutional layers remain frozen initially, and we stack additional dense layers (ReLU activation, batch normalization, dropout) on top. A softmax layer provides the final multi-class predictions.

4.2.2 Training Strategy

Following the standard approach from Section *Preprocessing*, we:

- Begin with a **Feature Extraction Phase**, training only the newly added dense layers.

- **Fine-Tune** the last few layers of MobileNetV2 (Conv2D, DepthwiseConv2D) to adapt to lung nodule features.
- Use a class-weighted loss (multiclass strategy described in Section *Preprocessing*) and early stopping to tackle data imbalance and overfitting.
- Optimize with Adam initially, then switch to RMSprop for fine-tuning.

Validation accuracy guides model selection, and the best-performing checkpoint is saved for subsequent evaluation.

4.3 Full Slice Binary Classification Model and Training

4.3.1 Model Architecture

Given the challenges in accurately classifying lung nodules using traditional CNN-based approaches, we explored an **autoencoder-based feature extraction** method. The proposed model consists of two main components:

- **Encoder:** A deep convolutional neural network that progressively compresses input full-slice CT images into a lower-dimensional latent space representation. The encoder is composed of several convolutional layers with batch normalization, LeakyReLU activations, and max-pooling to downsample the spatial dimensions while preserving meaningful patterns.
- **Decoder:** A symmetric counterpart of the encoder that reconstructs the original input image from the latent space representation. This ensures that the extracted features retain essential diagnostic information. The decoder consists of upsampling layers with transposed convolutions and batch normalization, gradually reconstructing the spatial details of the input CT scan.
- **Latent Space Representation:** The encoder outputs a compressed vector of dimension 64, which serves as a feature extractor. These latent embeddings are subsequently fed into a classifier to perform binary malignancy classification.

The inclusion of batch normalization and dropout regularization within the latent classifier helps prevent overfitting and enhances generalization. The final classifier consists of fully connected layers with ReLU activations, followed by a sigmoid activation for binary classification.

4.3.2 Training Strategy

The training process follows a **two-step approach**:

- **Step 1: Autoencoder Pretraining** – The autoencoder is initially trained in an unsupervised manner, optimizing for **image reconstruction**. The goal is to ensure that the encoder extracts meaningful feature representations while discarding irrelevant information. We use **Mean Absolute Error (MAE)** as the reconstruction loss function, with an **Adam optimizer** and an exponentially decaying learning rate.
- **Step 2: Classifier Training on Latent Features** – After training the autoencoder, we discard the decoder and use the encoder to extract features from the training and validation sets. These features are then used to train a binary classifier. The classifier is optimized using **binary cross-entropy loss** with class-weighted adjustments to mitigate class imbalance.
- **Fine-Tuning** – Once the classifier is trained, we fine-tune both the encoder and classifier jointly. The earlier layers of the encoder remain frozen to preserve low-level feature extraction, while the latter layers and classifier are updated during fine-tuning with a lower learning rate.

However, the reconstructed image shown in Fig. 7 demonstrates the limitations of the autoencoder. While the overall lung structure is preserved, fine-grained details, including nodule characteristics, are blurred or lost. This indicates that the encoder focuses on extracting high-level anatomical features while discarding noise and unstructured variations. Thus results suggest that while the autoencoder is capable of compressing lung CT slices into meaningful latent vectors, it may not optimally preserve nodule-specific features necessary for effective classification (see Figure 8 for confusion matrix). This

observation further motivated the exploration of **alternative feature extraction strategies**, such as attention mechanisms or hybrid CNN-transformer architectures, to enhance nodule-specific feature learning, but they didn't yield better results.

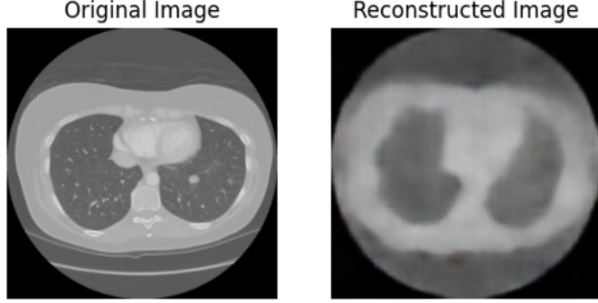


Figure 7: Reconstructed Image with AE

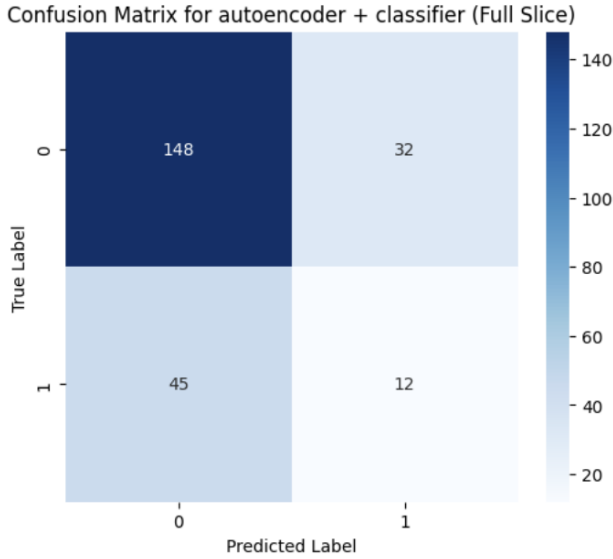


Figure 8: Confusion Matrix of the autoencoder and a custom classifier on the test set

4.4 Full Slice Multiclass Classification Model and Training

4.4.1 Model Architecture

For full-slice CT images (512×512 grayscale), we employ a custom CNN with four convolutional blocks (filter sizes from 32 to 256). Each block applies:

- A 3×3 convolution (ReLU activation, same padding, $L2$ regularization with $\lambda = 1 \times 10^{-4}$)
- Batch normalization
- 2×2 max pooling

A global average pooling layer replaces flattening, feeding into a Dense layer with 128 neurons (ReLU) and dropout (0.5). The final softmax layer produces a five-class distribution. To counter imbalance, we use a categorical focal loss ($\gamma = 2.0$, $\alpha = 0.25$). Optimization is performed with AdamW (lr= 1×10^{-3} , weight decay= 1×10^{-6}).

4.4.2 Training Strategy

We apply the usual dataset splits and class weighting from Section *Preprocessing*. Additionally, data augmentation (rotations, shifts, shear, zoom, flips) is used to improve generalization. Early stopping (patience=10, best weights restoration) and a learning rate scheduler (reducing lr by factor 0.1 after 3 epochs of no improvement, minimum lr= 1×10^{-5}) further refine training.

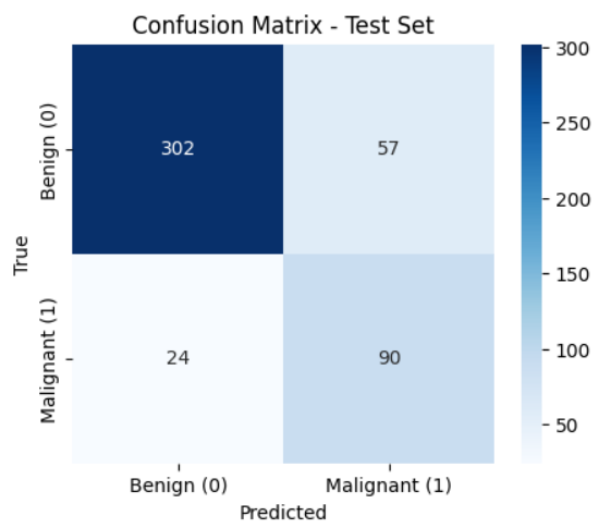


Figure 9: Confusion Matrix on the Test Set

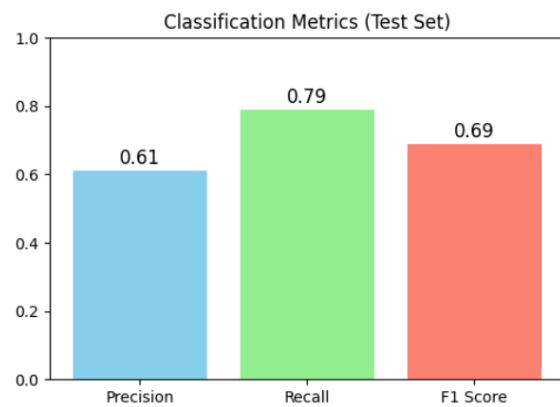


Figure 10: Classification Metrics on the Test Set

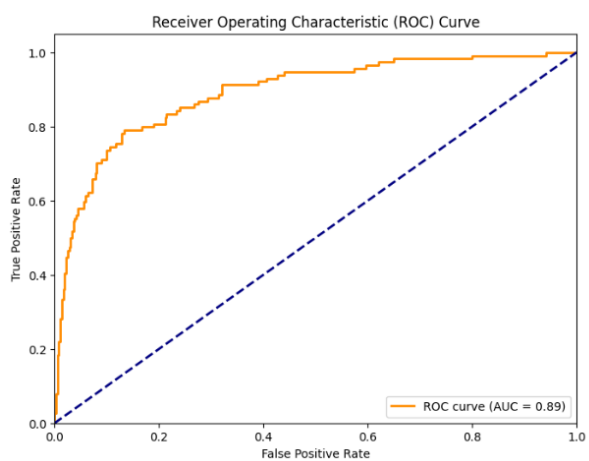


Figure 11: ROC Curve

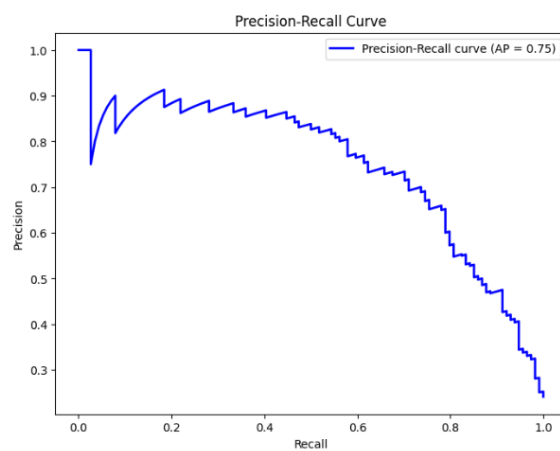


Figure 12: Precision-Recall Curve

5 Experiments

5.1 Simple Sequential Model for Zoomed Nodule Binary Classification

In this work, we developed a convolutional neural network (CNN) tailored for the binary classification of zoomed nodule images. The goal is to distinguish between benign and malignant tumors based on CT scan slices that focus on the nodule region. The following paragraphs detail the model architecture, the training process, and the rationale behind our design choices.

Model Architecture: The CNN model is constructed with a simple yet effective architecture:

- **Convolutional Blocks:** The network consists of three sequential convolutional blocks. Each block includes:
 - A Conv2D layer with ReLU activation to learn spatial features.
 - A MaxPooling2D layer to reduce spatial dimensions and control overfitting.
- **Flattening and Dense Layers:** After the convolutional blocks, the feature maps are flattened and passed through a fully connected (Dense) layer with 128 neurons and a ReLU activation.
- **Regularization:** A Dropout layer (with a dropout rate of 0.3) is applied to mitigate overfitting.
- **Output Layer:** The final layer is a single neuron with a sigmoid activation function, which outputs a probability for the malignant class.

This architecture is chosen for its simplicity, which helps reduce the risk of overfitting given the limited dataset size.

Training Process: The training process involves several key steps:

- **Data Preparation:**
 - *Image Loading:* Images are read in NRRD format, converted from 3D to 2D if necessary.

- *Preprocessing:* Each image is resized to 128×128 pixels and converted from grayscale to RGB to meet the model’s input requirements.

- **Class Balancing:** To counteract the class imbalance (with benign samples dominating), class weights are computed using the formula

$$w_c = \frac{1}{n_c} \times \frac{N}{2}$$

where n_c is the number of samples for class c and N is the total number of samples. This ensures that the minority class (malignant) has a stronger influence during training.

- **Cross-Validation:** A 5-fold stratified cross-validation is applied to ensure robust performance estimation. During cross-validation, early stopping and learning rate reduction callbacks monitor the validation loss to prevent overfitting.
- **Final Training:** After obtaining stable performance estimates, the model is retrained on the entire train+validation set using the final hyperparameters and class weights. The hold-out test set is used for the final evaluation.

Rationale Behind the Design Choices:

- **Simple Architecture:** The chosen CNN architecture is intentionally simple to accommodate the relatively small dataset and minimize overfitting.
- **ReLU and Pooling:** ReLU activations are computationally efficient and help mitigate the vanishing gradient problem, while max-pooling layers reduce dimensionality and capture dominant features.
- **Dropout:** A dropout rate of 0.3 is employed to further reduce overfitting by randomly deactivating neurons during training.
- **Class Weights:** The use of class weights addresses the severe imbalance between benign and malignant classes, ensuring that the minority class is adequately represented during training.

- **Cross-Validation and Callbacks:** Implementing cross-validation along with early stopping and learning rate reduction ensures that the training process is robust and that the model generalizes well to unseen data.

This systematic approach, from data preprocessing through model training, results in a robust model capable of accurately distinguishing between benign and malignant nodules in zoomed images.

5.2 MobileNetV2 for Zoomed Nodule Binary Classification

In this experiment, we employed a transfer learning approach using the MobileNetV2 architecture, pretrained on ImageNet, to classify zoomed nodule images as benign or malignant.

5.2.1 Data Preprocessing

The CT scan images were originally in NRRD format. We performed the following preprocessing steps:

- Extracted the relevant 2D slices from volumetric data.
- Resized images to **128×128 pixels** to ensure a consistent input shape.
- Converted grayscale images to **RGB format** for compatibility with MobileNetV2.

5.2.2 Model Setup

MobileNetV2 was selected for its efficiency and robust feature extraction capabilities. The model architecture consisted of:

- Base model initialized with **ImageNet weights**, removing the top classification layers.
- Freezing the first **100 layers** to retain low-level features while fine-tuning the deeper layers.
- Adding a **Global Average Pooling** layer, followed by a fully connected layer (128 neurons, ReLU activation).
- Applying a **Dropout** layer (0.3 rate) for regularization.

- A final neuron with a **sigmoid activation** for binary classification.

5.2.3 Training Strategy

To ensure robust performance, we implemented:

- **5-fold stratified cross-validation** to assess generalizability.
- **Class balancing** using weights calculated as:

$$w_c = \frac{1}{n_c} \times \frac{N}{2}, \quad (1)$$

where n_c is the number of samples in class c and N is the total dataset size.

- **Early stopping and learning rate scheduling** to prevent overfitting.

5.2.4 Results and Observations

MobileNetV2 provided competitive results, balancing accuracy and computational efficiency. However, fine-tuning required careful layer selection to optimize performance.

5.3 EfficientNetV2M for Full slice Multi Classification

5.3.1 Model Setup

The network accepts input images of size 512×512 with three channels; grayscale images are first converted to RGB and normalized to the $[0, 1]$ range. Data augmentation is implemented via a sequential pipeline incorporating RandAugment and GridMask to introduce controlled perturbations that improve generalization. The EfficientNetV2M model, pre-trained on ImageNet, is utilized as a feature extractor with its weights initially frozen. On top of this backbone, additional fully connected layers are appended—comprising dense layers with 512, 256, and 128 neurons, each followed by dropout and batch normalization—culminating in a softmax-activated output layer that provides a probability distribution over the five classes.

5.3.2 Training Strategy

The model is compiled using a categorical cross-entropy loss function and optimized with the Adam optimizer employing an exponential decay learning rate schedule (starting at 1×10^{-4} and decaying by a factor of 0.9 every 10,000 steps). To address class imbalance, personalized class weights are applied during training. The initial training phase runs for 20 epochs using a `tf.data` pipeline that integrates data augmentation and shuffling, with early stopping based on validation accuracy (patience of 10 epochs and restoration of best weights) to prevent overfitting. Subsequently, the model undergoes a fine-tuning phase under a mixed precision policy (`mixed_float16`), wherein selective unfreezing of convolutional layers (i.e., layers of type `Conv2D` and `DepthwiseConv2D`) is performed while maintaining the first 20 layers in a non-trainable state to preserve low-level features. This fine-tuning is conducted for up to 200 epochs, further refining the model’s performance for robust multi-class classification.

5.4 Models for Full Slice Binary Classification

In order to identify the best model for full slice binary classification, we conducted several experiments using different deep learning architectures.

- Initial CNN Models with Transfer Learning and Fine-Tuning:** Our first approach leveraged convolutional neural network (CNN) architectures with transfer learning and fine-tuning. Specifically, we tested EfficientNetV2S, MobileNetV2, and DenseNet121, given their strong performance in medical imaging classification tasks. However, all models exhibited similar performance, with an overall high accuracy but a critical weakness: the confusion matrix revealed a strong bias toward the negative class, failing to classify positive samples effectively. This limitation is particularly problematic in the medical field, where correctly identifying the presence of a nodule is crucial. (see Figure 13)

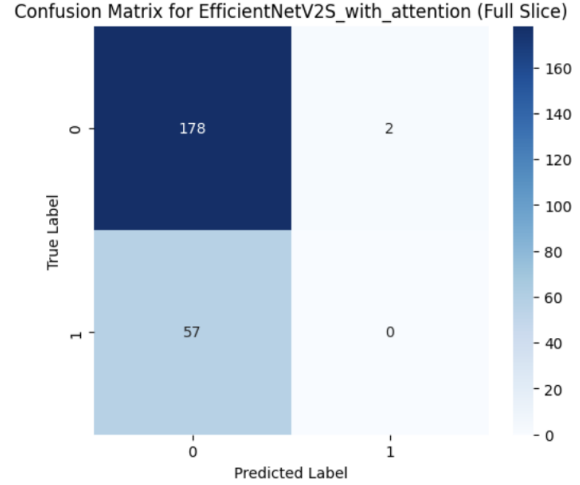


Figure 13: Confusion matrix of EfficientNetV2S on the test set

Incorporating Attention Mechanisms: To enhance feature extraction and better highlight diagnostically relevant regions, we introduced an attention mechanism in the classification head. Specifically, we experimented with a Squeeze-and-Excitation (SE) block to refine feature learning by adaptively re-weighting channels. Despite this modification, model performance did not show significant improvements, and the issue of poor sensitivity to the positive class remained.

Addressing Class Imbalance via Data Augmentation: Given the strong imbalance in the dataset, we investigated the effect of applying various data augmentation techniques specifically targeting the minority class (positive cases). However, these augmentation techniques failed to yield substantial performance gains. Instead, they introduced additional artifacts that did not contribute positively to model generalization.

Exploring Alternative Feature Extraction Approaches: Given the repeated failure of CNN models to properly extract relevant diagnostic features, we hypothesized that traditional convolutional networks might not be the optimal solution for feature representation in this domain. To address this, we implemented an autoencoder-based feature extractor followed by a custom classifier.

- Autoencoder Feature Learning:** The encoder was designed to extract compressed feature representations, incorporating either:

- **Attention (SE Blocks):** To focus on diagnostically significant regions.
- **Global Average Pooling (GAP):** To replace fully connected layers and reduce overfitting.

The extracted representations from the autoencoder were then used as input for a fully connected classifier. See results in Table 2

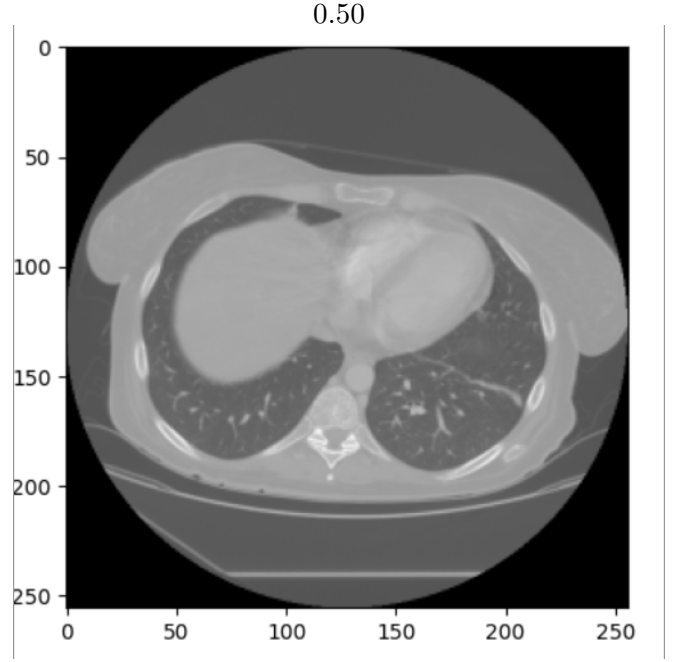


Figure 14: Original CT slice

5.4.1 Analysis through Grad-CAMs:

To further investigate whether CNN-based models were effectively learning discriminative features for nodule classification, we applied **Gradient-weighted Class Activation Mapping (Grad-CAM)**. Grad-CAM is a visualization technique that highlights the regions of an image that contribute most to the model’s predictions by computing the gradient of the output class score with respect to the feature maps of the last convolutional layer. The gradients are then globally averaged to generate a weighted activation map, which is overlaid onto the original image to interpret where the model is focusing.

In our experiments, we used Grad-CAM to analyze the feature attribution of the **DenseNet121**-based classifier. The resulting heatmap visualizations are presented in Fig. 15, comparing the original CT slice with the Grad-CAM activation.

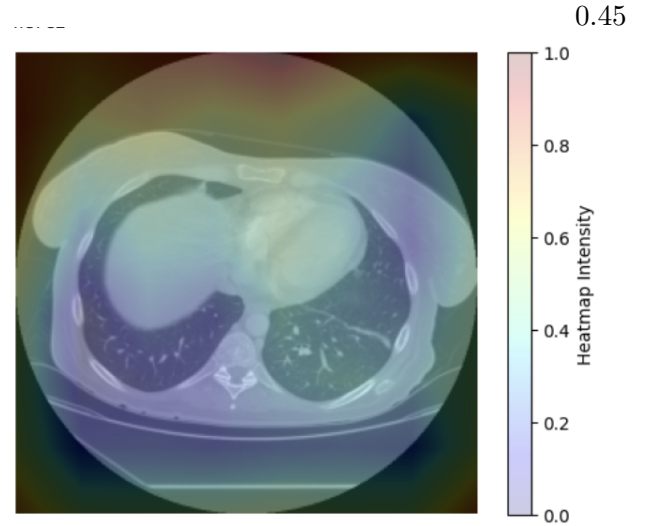


Figure 15: Grad-CAM heatmap overlay (DenseNet121)

Figure 16: Visualization of model attention using Grad-CAM.

The heatmap in Fig. 15 shows that the **DenseNet121** model fails to consistently focus on the lung nodules. Instead, its activations are spread across irrelevant regions, including the lung periphery and non-informative background areas. This suggests that the model struggles to localize nodules effectively, leading to misclassifications and re-

enforcing the hypothesis that CNNs alone may not be optimal for this task.

This analysis motivated the exploration of alternative strategies, such as incorporating **attention mechanisms** to enhance feature learning or leveraging **autoencoder-based** approaches for improved feature extraction. These modifications aimed to ensure that the model prioritizes diagnostically relevant regions, rather than distributing attention across the entire CT scan.

5.4.2 Final Evaluation and Conclusion:

While the autoencoder-based approach resulted in a slightly lower overall accuracy compared to the CNN models, it produced a significantly improved confusion matrix, correctly identifying more positive cases. This trade-off suggests that an explicit feature extraction phase, rather than direct classification through CNNs, may be more effective for this medical imaging task.

A potential future improvement involves incorporating a two-stage pipeline that integrates nodule localization with object detection and patch-based classification. A dedicated object detection model, such as YOLO or Faster R-CNN, can be trained using an annotated dataset with bounding boxes to first identify the presence and location of nodules within full-slice CT scans. Once the nodule is localized, the extracted region of interest (ROI) can be passed through a specialized classifier for malignancy prediction. This classification step could leverage CNN-based models or hybrid architectures incorporating transformers, such as Swin Transformers or Vision Transformers (ViTs), to better capture long-range dependencies in the extracted nodule patches. Such a hybrid approach could significantly improve diagnostic accuracy by ensuring that only the relevant regions are analyzed while allowing the classifier to focus on fine-grained features of the nodules rather than processing the entire full-slice CT scan.

6 Results

Table 1: Binary Classification (Benign, Malignant) on Nodules

Model Name	Accuracy	Precision	Recall	F1-score
DenseNet	0.83 ± 0.034	0.61	0.79	0.69
MobileNet	0.69 ± 0.042	0.90	0.67	0.77
Sequential	0.77 ± 0.038	0.86	0.91	0.89

Table 2: Binary Classification (Benign, Malignant) on Full Slice Model

Model Name	Accuracy	Precision	Recall	AUC
EfficientNetV2S	0.7595 ± 0.012	0.6988	0.7595	0.5623
EfficientNetV2S + Attention	0.7511 ± 0.010	0.5753	0.7511	0.5072
MobileNetV2	0.7595 ± 0.011	0.5768	0.7595	0.5884
DenseNet121	0.7595 ± 0.014	0.5768	0.7595	0.6401
Autoencoder + Classifier	0.6751 ± 0.015	0.6480	0.6751	0.5437

Table 3: Multiclass Classification (1,2,3,4,5) on Nodules

Model Name	Accuracy	Precision	Recall	F1-score
EfficientNetV2B0	0.47 ± 0.063	0.49	0.47	0.48

Table 4: Multiclass Classification (1,2,3,4,5) on Full

Model Name	Accuracy	Precision	Recall	F1-score
EfficientNetV2M	0.31 $\pm$ 0.033	0.35	0.26	0.26
EfficientNetV2B0	0.27 $\pm$ 0.056	0.35	0.27	0.29
Sequential	0.46 $\pm$ 0.064	0.19	0.22	0.17

6.1 Confidence Estimates

To evaluate our models, we considered not only the illustrated metrics but also additional parameters:

- **Confidence Intervals:** Accuracy, sensitivity (or recall), precision, and F1-score provide insights into predictive performance, evaluating correctness (accuracy), the ability to identify true positives (sensitivity), and true negatives (specificity).
- However, reporting test values alone is not sufficient to assess model robustness. Thus, we

computed confidence intervals on accuracy using the following equation:

$$ACC_{test} \pm z \sqrt{\frac{1}{n} ACC_{test}(1 - ACC_{test})} \quad (2)$$

A confidence level of 95% was used.

- **AUC and ROC Curve:** In addition to standard metrics, we analyzed the Area Under the Curve (AUC) and the Receiver Operating Characteristic (ROC) curve, along with the Precision-Recall curve, to gain deeper insights into the model’s classification performance.

7 Conclusions

The development of classification models for tumor detection presented several challenges, particularly in addressing dataset imbalance. To mitigate this issue, the use of class weighting and data augmentation proved to be crucial.

For full-slice multiclass classification, the complexity of the task increased significantly due to the high similarity between certain classes and the relatively small spatial region occupied by the tumor compared to the overall chest area. This made distinguishing between tumor categories more difficult, impacting classification performance.

Conversely, in the zoomed nodule classification, the clearer visibility of the tumor led to improved

model performance. This effect was particularly evident in the binary classification task, where inter-class differences were less significant. As a result, even simple sequential models, without augmentation, achieved highly satisfactory results, demonstrating that when the tumor is more prominently represented in the input data, classification accuracy improves significantly.

These findings highlight the importance of pre-processing strategies and dataset representation in medical image classification, emphasizing the need for task-specific adaptations to optimize predictive performance.